ECE 329 | Fields and Waves I | Spring 19

University of Illinois | Chen, Goddard, Zhao

Exam 3

Wednesday, April 17, 2019 — 8:30-9:45 PM

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the boxes below.

Name:	Solutions
NetID:	
Section:	12 Noon 1 PM 2 PM

If you want your exam returned to a **different** section, write it here:

Special Return to Section:	12 Noon 1 PM 2 PM

This is a closed book exam and calculators/electronic devices are not allowed. You are allowed THREE double sided 3"x5" index card of notes. Please show all your work and make sure to include your reasoning for each answer. All answers should include **units** wherever appropriate. The exam is double sided. You may use the back of the exam as scratch paper.

Problem 1 (25 points)	
Problem 2 (20 points)	
Problem 3 (15 points)	
Problem 4 (25 points)	
Problem 5 (15 points)	
TOTAL (100 points)	

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

Because calculators are not allowed, please use the following expressions for physical constants:

$$c = 3 \times 10^8 \left[\frac{m}{s}\right]$$

$$\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \left[\frac{F}{m}\right]$$

$$\mu_0 = 4\pi \times 10^{-7} \left[\frac{H}{m}\right]$$

$$\eta_0 = 120\pi \ [\Omega]$$

and leave your answers in the form of fractions rather than decimals.

A Table for Your Reference

23 Imperfect dielectrics, good conductors

	Condition	β	α	$\lvert\eta\rvert$	τ	$\lambda = \frac{2\pi}{\beta}$	$\delta = \frac{1}{\alpha}$
Perfect dielectric	$\sigma = 0$	$\omega\sqrt{\epsilon\mu}$	0	$\sqrt{\frac{\mu}{\epsilon}}$	0	$\frac{2\pi}{\omega\sqrt{\epsilon\mu}}$	∞
Imperfect dielectric	$\frac{\sigma}{\omega\epsilon} \ll 1$	$\sim \omega\sqrt{\epsilon\mu}$	$\beta\frac{1}{2}\frac{\sigma}{\omega\epsilon} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$	$\sim \sqrt{\frac{\mu}{\epsilon}}$	$\sim \frac{\sigma}{2\omega\epsilon}$	$\sim \frac{2\pi}{\omega\sqrt{\epsilon\mu}}$	$\frac{2}{\sigma}\sqrt{\frac{\epsilon}{\mu}}$
Good conductor	$\frac{\sigma}{\omega\epsilon} \gg 1$	$\sim \sqrt{\pi f\mu\sigma}$	$\sim \sqrt{\pi f\mu\sigma}$	$\sqrt{\frac{\omega\mu}{\sigma}}$	$45°$	$\sim \frac{2\pi}{\sqrt{\pi f\mu\sigma}}$	$\sim \frac{1}{\sqrt{\pi f\mu\sigma}}$
Perfect conductor	$\sigma = \infty$	∞	∞	0	-	0	0

- The table above summarizes TEM wave parameters in homogeneous conducting media where the propagation velocity

$$v_p = \frac{\omega}{\beta}$$

x-polarized phasor

$$\tilde{\mathbf{E}} = \hat{x}E_o e^{\mp\alpha z} e^{\mp j\beta z}$$

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

1. (25 points) In a general media, plane TEM wave parameters γ and η satisfy:

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \text{ and } \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}. \quad \text{Thus, } \gamma\eta = j\omega\mu \text{ and } \frac{\gamma}{\eta} = \sigma + j\omega\epsilon$$

A plane TEM wave propagating with $\omega = 2\pi \cdot 10^7$ [rad/s] in a material medium with $\mu = 2\mu_0$ has an electric field phasor:

$$\tilde{\mathbf{E}} = 20e^{-(1+j\sqrt{3})y}\hat{z} \text{ V/m}$$

Determine the following properties of the wave:

a) (2 pts) The propagation constant γ expressed as $\alpha + j\beta$.

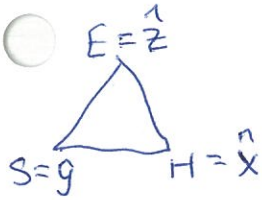
Your Answer (include appropriate units): $\alpha =$ 1 Np/m $\beta =$ $\sqrt{3}$ rad/m

b) (7 pts) The impedance η expressed as $|\eta|e^{j\tau}$. **Hint:** $1 + j\sqrt{3} = 2e^{j60°}$

$\eta = \frac{j\omega\mu}{\gamma} = j\frac{2\pi\cdot 10^7 \cdot 2 \cdot 4\pi\cdot 10^{-7}}{2e^{j60}} = e^{j90-j60}\cdot 8\pi^2 = 8\pi^2 e^{j30°}$ ($\mu = 2\mu_0$)

Your Answer (include appropriate units): $|\eta| =$ $8\pi^2\,\Omega$ $\tau =$ $30°$

c) (8 pts) The magnetic field phasor $\tilde{\mathbf{H}}(y)$



$\tilde{H} = \frac{\tilde{E}}{\eta} = \frac{20}{8\pi^2} e^{-(1+j\sqrt{3})y} e^{-j30°}\hat{x}$

Your Answer (include direction and appropriate units): $\tilde{\mathbf{H}}(y) = \frac{5}{2\pi^2} e^{-(1+j\sqrt{3})y - j30°}\hat{x}$ A/m

d) (6 pts) The loss tangent $\frac{\sigma}{\omega\epsilon}$. **Hint:** $cos(30°) = sin(60°) = \frac{\sqrt{3}}{2}$, $cos(60°) = sin(30°) = \frac{1}{2}$.

$\frac{\bar{\gamma}}{\eta} = \sigma + j\omega\epsilon \Rightarrow \frac{\sigma}{\omega\epsilon} = \frac{Re[\frac{\bar{\gamma}}{\eta}]}{Im[\frac{\bar{\gamma}}{\eta}]}$. $\frac{\bar{\gamma}}{\eta} = \frac{2e^{j60°}}{4\pi^2 e^{j30°}} = \frac{1}{2\pi^2}e^{j30} = \frac{1}{2\pi^2}\left(\frac{\sqrt{3}}{2} + j\frac{1}{2}\right)$

$\Rightarrow \frac{\sigma}{\omega\epsilon} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$

Your Answer (include appropriate units): $\frac{\sigma}{\omega\epsilon} =$ $\sqrt{3}$

e) (2 pts) This material would be best classified as a (CIRCLE ONE):
(A) perfect dielectric, (B) imperfect dielectric,
(C) in between an inperfect dielectric and a good conductor,
(D) good conductor, (E) perfect conductor.

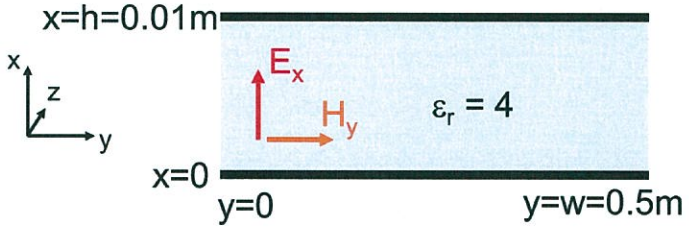
because $\frac{\sigma}{\omega\epsilon}$ is not $\gg 1$ for good conductor or $\ll 1$ for imperfect dielectric

Can also tell its in between because $\alpha \ll \beta$ for imperfect dielectric, $\alpha \approx \beta$ for good conductor

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

Can also tell because $\tau \neq 0°$ imperfect dielectric and $\tau \neq 45°$ good conductor

2. (20 points) A parallel plate transmission line has a length $l = 15$ [m]. The cross-section has a width of $w = 0.5$ [m] and a height of $h = 0.01$ [m] and is filled with a perfect dielectric material $\epsilon_r = 4$ that is non-magnetic, $\mu_r = 1$. Note therefore that the intrinsic impedance is $\eta = \frac{1}{2}\eta_0 = 60\pi$ [Ω].



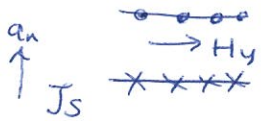
a) (5 pts) Find the characteristic impedance Z_0.

$$Z_0 = \frac{V}{I} = \frac{E \cdot h}{H \cdot w} = \eta \cdot \frac{h}{w} = 60\pi \cdot \frac{0.01}{0.5} = 1.2\pi$$

Alternatively, $Z_0 = \sqrt{\frac{\mathcal{L}}{\mathcal{C}}} = \sqrt{\frac{\mu/(Wd)}{\varepsilon W/d}} = \eta \frac{d}{w} = 60\pi \cdot \frac{0.01}{0.5} = 1.2\pi$

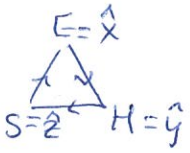
Your Answer (include appropriate units): $Z_0 =$ 1.2π Ω

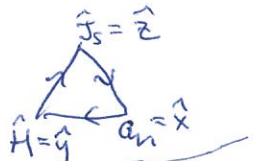
b) (8 pts) A current flows along the bottom plate is $I = 2\delta(t - \frac{z}{v})$ [A] in the positive z-direction and returns on the top plate. Find the corresponding electric field $\mathbf{E}(z,t)$.



$$H_y = 2 \cdot \frac{1}{2} J_s \times a_n = |J_s| = \frac{I}{w}$$

↑ top and bottom plates





$$E_x(z,t) = \eta H(z,t) = \frac{60\pi \; 2\delta(t - \frac{z}{v})}{0.5} = 240\pi\delta(t - \frac{z}{v})$$

Alternatively, $V = Z_0 \cdot I = 2.4\pi\delta(t - \frac{z}{v})$

$E_x = \frac{V}{h} = 240\pi\delta(t - \frac{z}{v})$

Your Answer (include direction and appropriate units): $\mathbf{E}(z,t) =$ $240\pi\delta(t - \frac{z}{v})\hat{x}$ V/m

c) (7 pts) If instead the electrical field phasor for some frequency ω_0 is $\tilde{\mathbf{E}} = 240\pi e^{-j\beta z}\hat{x}$ [V/m], what is the time averaged power crossing this transverse cross section?

$$\langle \vec{S} \rangle = \frac{1}{2} \mathrm{Re}[\tilde{E} \times \tilde{H}^*] = \frac{1}{2} \mathrm{Re}\left[\frac{1}{\eta}|\tilde{E}|^2 \hat{z}\right] = \frac{(240\pi)^2}{120\pi} = 2 \cdot (240\pi)$$

$$\langle P \rangle = \langle \vec{S} \rangle \cdot \mathrm{Area} = 480\pi \cdot 0.01 \cdot 0.5 = 2.4\pi \text{ W}$$

Alternatively, $\langle P \rangle = \frac{1}{2}\mathrm{Re}[\tilde{V} \cdot \tilde{I}^*] = \frac{1}{2}\left[\frac{1}{Z_0}|\tilde{V}|^2\right] = \frac{1}{2.4\pi}(2.4\pi)^2 = 2.4\pi$ W

Your Answer (include appropriate units): $< P > =$ 2.4π W

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

3. (15 points) All parts of this problem are independent, i.e., you do not necessarily need the correct solution in the previous part(s) to find the correct solution. Consider an incident TEM wave propagating in free space, where the electric field is

$$\mathbf{E}_i(z,t) = \cos(\omega t + \beta x)\hat{y} + \sin(\omega t + \beta x)\hat{z}\,[V/m]$$

a) (4 points) What is the polarization of the incident wave?

i. Linear
ii. LHCP
iii. RHCP
iv. Elliptical
v. None of the above

Equal magnitude and out of phase by $\frac{\pi}{2}$ $\Rightarrow$ Circular
Direction of propagation: $-\hat{x}$
Leading x Lagging: $\hat{y} \times \hat{z} = +\hat{x} \Rightarrow$ Left-hand

b) (7 points) At $x = 0$, the wave strikes the boundary between the infinite half space of free space and the infinite half space consisting of a perfect electrical conductor (PEC). What is the electric field of the reflected wave?

$$\Gamma = \frac{0 - \eta_0}{0 + \eta_0} = -1$$

Reflected wave in $+\hat{x}$ direction

Your Answer (include direction and appropriate units):

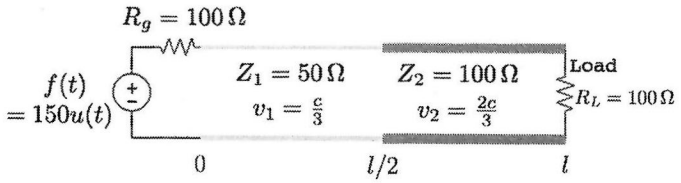
$\mathbf{E}_r(z,t) =$ $-\cos(\omega t - \beta x)\,\hat{y} - \sin(\omega t - \beta x)\,\hat{z}$ $\left[\frac{V}{m}\right]$

c) (4 points) What best describes the total wave in the free space region?

i. All traveling wave
ii. All standing wave
iii. Some traveling wave, some standing wave
iv. Some traveling wave, some evanescent wave
v. None of the above

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

4. (25 points) Except for (e) and (f), all parts of this problem are independent, i.e., you do not necessarily need the correct solution in the previous part(s) to find the correct solution. Consider the circuit below where two transmission lines of equal lengths are joined directly at a distance $\frac{l}{2}$ away from the generator. Let $Z_1 = 50\,\Omega$, $v_1 = \frac{c}{3} \approx 1 \times 10^8\,\frac{\text{m}}{\text{s}}$, $l = 40\,\text{m}$, and the voltage source $f(t) = 150u(t)\,\text{V}$.



a) (3 points) What are the voltage and current **reflection** coefficients $\Gamma_{12,V}$ and $\Gamma_{12,I}$ at the junction $z = \frac{l}{2}$ for the wave moving to the right?

$$\Gamma_{12,V} = \frac{100 - 50}{100 + 50} = \frac{1}{3}$$

Your Answer (include appropriate units): $\Gamma_{12,V} =$ $\frac{1}{3}$ $\Gamma_{12,I} =$ $\frac{-1}{3}$

b) (3 points) What are the voltage and current **transmission** coefficients $\tau_{12,V}$ and $\tau_{12,I}$ at the junction $z = \frac{l}{2}$ for the wave moving to the right?

$$\tau_{12,V} = 1 + \Gamma_{12,V} = \frac{4}{3} \qquad \tau_{12,I} = 1 + \Gamma_{12,I} = \frac{2}{3}$$

Your Answer (include appropriate units): $\tau_{12,V} =$ $\frac{4}{3}$ $\tau_{12,I} =$ $\frac{2}{3}$

c) (2 points) What are the voltage and current **reflection** coefficients $\Gamma_{L,V}$ and $\Gamma_{L,I}$ at the load for the wave moving to the right?

$$\Gamma_{L,V} = \frac{100 - 100}{100 + 100} = 0$$

Your Answer (include appropriate units): $\Gamma_{L,V} =$ 0 $\Gamma_{L,I} =$ 0

d) (2 points) What are the voltage and current **reflection** coefficients $\Gamma_{g,V}$ and $\Gamma_{g,I}$ at the generator for the wave moving to the left?

$$\Gamma_{g,V} = \frac{100 - 50}{100 + 50} = \frac{1}{3}$$

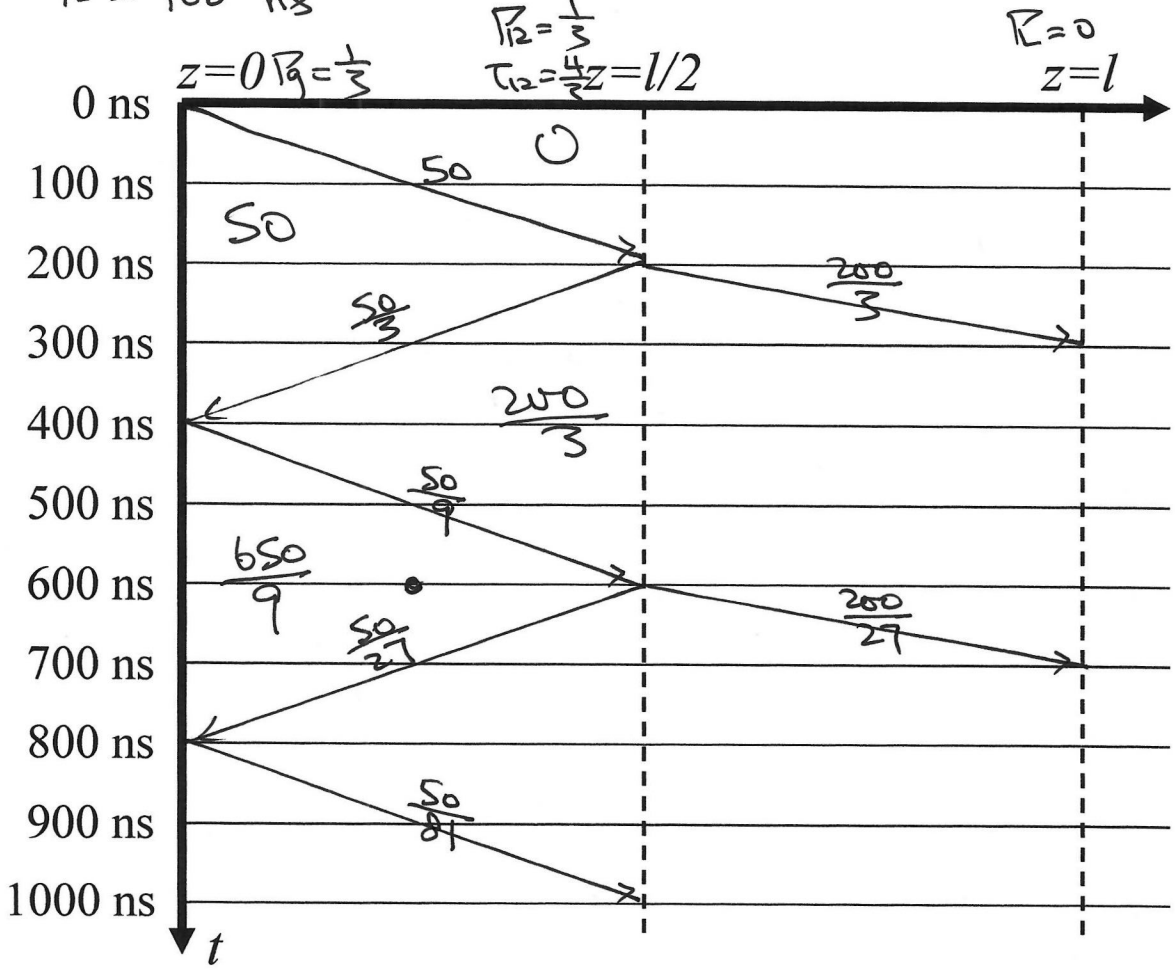
Your Answer (include appropriate units): $\Gamma_{g,V} =$ $\frac{1}{3}$ $\Gamma_{g,I} =$ $\frac{-1}{3}$

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

e) (10 pts) Complete the **voltage** bounce diagram below for the transmission line system from $t = 0$ to $t = 1\,\mu$s. Be sure to label all values clearly. $\Gamma_g = \frac{50}{100+50} = \frac{1}{3}$

$T_1 = \frac{20\text{ m}}{1\times 10^8\text{ m/s}} = 200\text{ ns}$

$T_2 = 100\text{ ns}$



f) (2 pts) Use the diagram to find $V(z = 10\,\text{m}, t = 650\,\text{ns})$.

Your Answer (include appropriate units):

$V(z = 10\,\text{m}, t = 650\,\text{ns}) =$ $\frac{650}{9}$ [V]

g) (3 pts) What is the steady state voltage across the load.

TL behaves like a wire at DC steady-state

Your Answer (include appropriate units):

$V(z = 40\,\text{m}, t = \infty\,\text{ns}) =$ 75 [V]

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.

5. (15 points) A lossless transmission line of length $l = 100$ [m] and velocity $v = 2 \times 10^8$ [m/s] supports resonant oscillations. In co-sinusoidal steady state, the voltage phasor of these oscillations is measured to be $\tilde{V}_{in} = 0$ [V] at the input and the current phasor at the load is measured to be $\tilde{I}_L = 0$ [A].

a) (5 pts) What are the input impedance Z_{in} and the load impedance Z_L for these resonances?

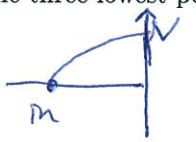
$Z_{in} = \frac{\tilde{V}_{in}}{\tilde{I}_{in}} = \frac{0}{\tilde{I}_{in}} = 0$ because $\tilde{I}_{in} \neq 0$

$Z_L = \frac{\tilde{V}_L}{\tilde{I}_L} = \frac{\tilde{V}_L}{0} = \infty$ because $\tilde{V}_L \neq 0$

Your Answer (include appropriate units):
$Z_{in} =$ 0 Ω $Z_L =$ ∞ Ω

b) (10 pts) Determine the three lowest possible frequencies for resonant oscillations.

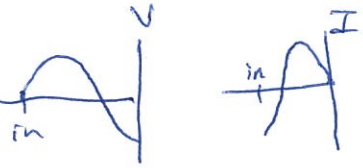
Lowest:



$\ell = \frac{1}{4}\lambda \Rightarrow \lambda = 4\ell$

$f = \frac{v}{\lambda} = \frac{2 \cdot 10^8}{400} = 0.5\,\text{MHz}$

Next lowest



$\ell = \frac{3}{4}\lambda \Rightarrow \lambda = \frac{4\ell}{3}$

$f = \frac{v}{\lambda} = \frac{2 \cdot 10^8}{400/3} = 1.5\,\text{MHz}$

Third:



$\ell = \frac{5}{4}\lambda \Rightarrow \lambda = \frac{4\ell}{5}$

$f = \frac{v}{\lambda} = \frac{2 \cdot 10^8}{400/5} = 2.5\,\text{MHz}$

Your Answer (include appropriate units):
$f_1 =$ 0.5 MHz, $f_2 =$ 1.5 MHz, $f_3 =$ 2.5 MHz

© 2019 Lynford Goddard. All rights reserved. Redistributing without permission is prohibited.